Polygenic Risk Score Portability Across Ancestries

BIOT 6900 · AI-Powered Biotech Workflows · Module 6, Part 3 (Computational Lab 6) · Fall 2026

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1. Summary

I used a published Alzheimer's disease (AD) polygenic risk score (PGS Catalog PGS000334), developed using European data, on real genetic data from 2,336 people in the 1000 Genomes Project across five population groups. The score included 22 genetic variants, including the APOE SNPs rs429358 and rs7412. All 22 variants were found and matched correctly, with 13 variants flipped and none removed. The score performed best in the European (EUR) group, with an AUC of 0.672, but its performance was lower in other groups. In the African (AFR) group, the AUC dropped to 0.493, which is close to random prediction. The score also showed large differences between populations. Using the same top 10% cutoff, 28.0% of African individuals were flagged, compared with only 2.2% of East Asian (EAS) individuals. When the score was combined with age, the cross-validated AUC increased from 0.562 to 0.618. This shows that the score improved prediction to some extent, but the overall improvement was still modest.

Important: 1000 Genomes has no disease labels, so the phenotype is **simulated**. The genotypes, published weights, ancestry labels, and PRS distributions are real; the AUC and calibration results use the simulated label and are **illustrative only**.

2. Data and methods

Inputs: The analysis used four files: genotypes.csv containing ALT-allele dosages (0, 1, or 2), variants.csv containing REF and ALT alleles from the same genome build, score.csv containing the PGS000334 effect alleles, other alleles, and their log-odds weights, and meta.csv containing ancestry, age, and simulated phenotype data. The dataset included 500 European (EUR), 500 African (AFR), 500 East Asian (EAS), 489 South Asian (SAS), and 347 American (AMR) individuals.

Workflow:

1**. Data harmonization:** Matched effect alleles with ALT/REF alleles, flipped dosage when needed (2 − dosage), and removed unmatched variants.

2. **PRS calculation:** Calculated PRS using genotype dosages and weights, then standardized scores across the whole dataset.

3**. Score distribution:** Compared PRS histograms across ancestry groups and calculated the percentage above the pooled 90th-percentile cutoff.

4. **AUC analysis:** Measured AUC for the full dataset and each ancestry group to evaluate prediction performance.

5. **Risk prediction:** Trained a logistic regression model on EUR data and applied it to all groups. Used five-bin calibration curves to compare predicted and observed risks.

6**. Model comparison:** Used five-fold cross-validation to compare AUC for age alone versus age combined with PRS.

Proof of real data (Section 2 output)

```
source      : REAL
people      : 2336
score size  : 22 variants
by ancestry : {'AFR': np.int64(500), 'EAS': np.int64(500), 'EUR': np.int64(500), 'SAS': np.int64(489), 'AMR': np.int64(347)}
  genotypes.csv: sha256 9cf41a1f
  variants.csv: sha256 f7a87d3f
  score.csv: sha256 7f80e8ff
  meta.csv: sha256 6cc55b6e
```

3. Results

3.1 Allele harmonization

All 22 variants from the score were found in the genotype data. Out of these, 9 were used directly because the effect allele matched the ALT allele, and 13 were flipped because the effect allele matched the REF allele. No variants were removed due to allele mismatch. This means the full score was used, and no strand or genome build issues were identified. Flipping the alleles was important because, without it, more than half of the variants would have counted the wrong allele and could have given incorrect risk scores.

3.2–3.3 PRS distribution and a single cutoff

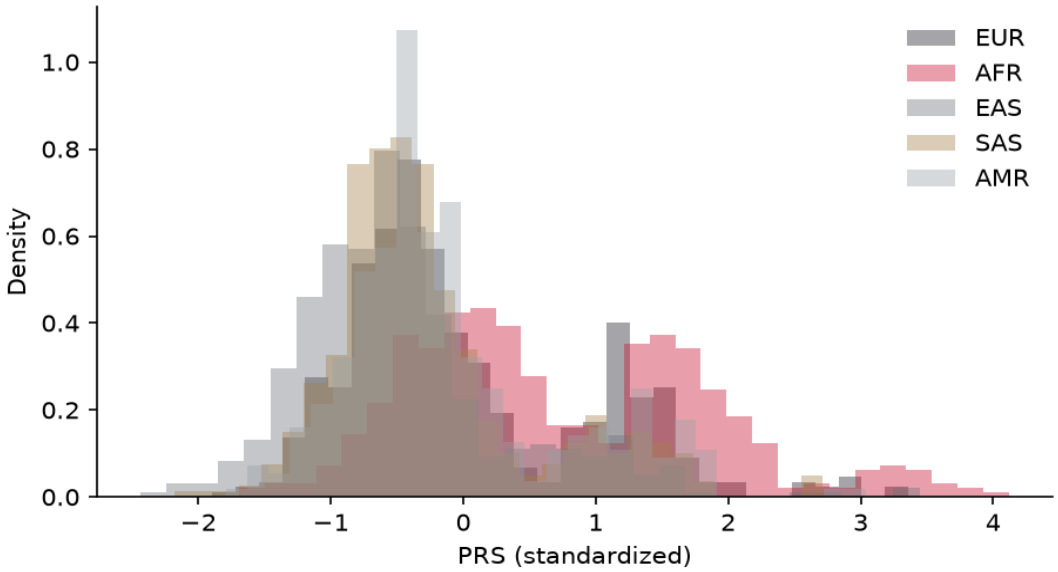


Figure 1. Standardized PRS by ancestry (one pooled standardization). The groups are centred at different places, and the clumps (around 0, 1.3 and 3) are most likely the dosage groups of the large-effect APOE variant, which carries much of this score.

	EUR	AFR	EAS	SAS	AMR
Mean PRS (standardized)	+0.036	+0.744	−0.472	−0.245	−0.099
% above pooled top-10% cutoff (1.493)	7.8%	28.0%	2.2%	4.3%	6.3%

Table 1. Mean PRS and the share of each group above one fixed cutoff. This uses no phenotype, only real genotypes and real weights.

3.4 Discrimination (AUC)

	Overall	EUR	AFR	EAS	SAS	AMR
AUC	0.599	0.672	0.493	0.575	0.611	0.555

Table 2. AUC overall and within each ancestry (simulated phenotype). Groups are 347–500 people with no confidence intervals, so differences of a few hundredths are within noise; the EUR vs. AFR gap is not.

3.5 Calibration

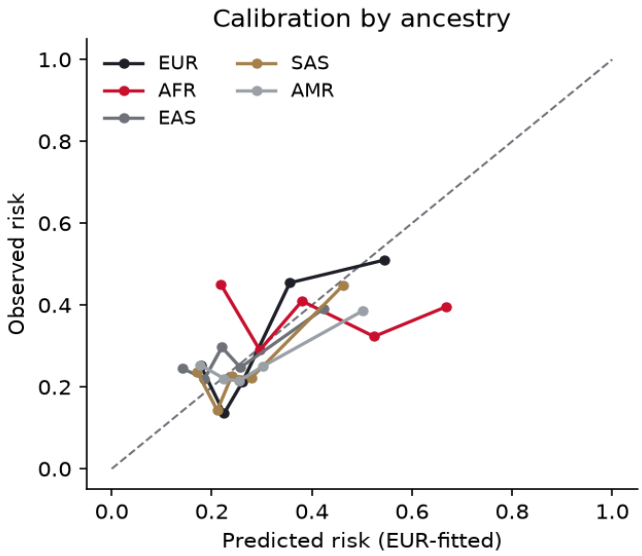


Figure 2. Calibration of the EUR-fitted risk model, applied to every ancestry. Dashed line = perfect calibration.

EUR follows the diagonal line fairly well, as expected because the model was trained on this group. SAS also shows a reasonable match. AMR and EAS show weaker agreement between predicted and actual risk. AFR shows the poorest results: predicted risk ranges from about 0.22 to 0.67, but the actual risk stays around 0.3–0.45 without a clear trend. The lowest predicted risk bin has an actual risk of about 0.45, while the highest has about 0.40.

3.6 Incremental value over age

Cross-validated AUC: **age only 0.562 → age + PRS 0.618** (+0.056). The PRS adds information beyond age, but the combined model is still modest, which is the honest expectation for a common-disease PRS.

4. Interpretation

4.1 Why the European-derived score loses accuracy elsewhere

A PRS usually does not use the actual disease-causing variants. Instead, it uses nearby SNPs identified in European studies that are linked to the causal variants through linkage disequilibrium (LD). These links can be different in other ancestry groups. African populations generally have shorter LD blocks, so European SNPs may not be strongly linked to the actual causal variants, making their weights less accurate. Also, allele frequencies differ between populations, which can shift the average PRS from −0.47 in EAS to +0.74 in AFR without showing a real difference in disease risk. This does not mean the disease biology is different between groups; it means the score was mainly developed using European data. This is reflected in the results, with the highest AUC in EUR (0.672), followed by SAS (0.611), and lower performance in AMR, EAS, and AFR.

4.2 One threshold, many groups

Using the same top 10% cutoff for all groups is not fair. It flags 28.0% of AFR individuals as high-risk, compared with only 2.2% of EAS individuals, even though the score performs no better than chance in AFR (AUC 0.493). If used in hospitals, African-ancestry patients could face unnecessary tests, extra costs, and anxiety, while East and South Asian patients with higher risk might be missed. These differences are mainly due to allele frequency and LD differences, not actual disease risk. At minimum, separate cutoffs should be considered for each ancestry group and validated using real patient data before clinical use. Ethically, this means a fixed cutoff would spread the benefits and harms of screening unequally by ancestry, deepening existing health disparities, so a score like this should not guide care, trial access, or insurance decisions until it is validated in every group it is used on.

4.3 Discrimination vs. calibration

Discrimination (AUC) checks whether the score ranks patients with the disease higher than those without it, while calibration checks whether the predicted risk matches the actual risk. Both results were good for EUR, while AFR showed poor performance in both, with an AUC close to chance. EAS and AMR had moderate AUC values of 0.575 and 0.555, but their calibration curves showed that the predicted risks did not match the actual risks. In AFR, the predicted risk was the highest because the PRS was shifted upwards, but the actual risk did not follow the same pattern. This means the model could give African patients incorrect risk estimates with high confidence. SAS showed better results, with an AUC of 0.611 and calibration close to the diagonal, suggesting reasonable performance. Overall, both AUC and calibration are important and should be checked separately for each ancestry group, rather than only for the combined population.

4.4 Clinical utility and limits

Where it could help: For adults of European ancestry, this score can be used along with age, family history, and APOE genotype to estimate risk. It may help select people for prevention trials or identify those who need earlier cognitive monitoring. The improvement over age alone (AUC increased from 0.562 to 0.618) is real but small. So, it can help assess risk, but it cannot be used to diagnose Alzheimer's disease.

Where it should not be used: It should not be used to diagnose Alzheimer's disease or predict an individual's risk on its own. It should not be used for non-European patients without proper validation and adjustment for each ancestry group. A single cutoff for all populations should also be avoided. The score should not be used to make decisions that affect access to healthcare or insurance.

What is needed before deployment: Use large and diverse patient groups with confirmed Alzheimer's disease cases and controls, preferably supported by amyloid or tau biomarkers. Include data from ADSP, All of Us, UK Biobank, and African- and Asian-ancestry AD studies. Check AUC and calibration separately for each ancestry group, including confidence intervals. Improve the score using multi-ancestry methods such as PRS-CSx and adjust it for each group. Finally, check whether using the score improves patient outcomes. Since this study used simulated disease data, its AUC and calibration results are not clinical evidence.

4.5 Artifact versus biology (APOE in AFR)

The score performs close to chance in AFR partly because the APOE ε4 proxy is almost absent in the simulated African-ancestry data, so the score cannot properly capture its effect. This is due to how the dataset and simulated disease outcomes were created, not human biology. In real populations, APOE ε4 is a risk factor for Alzheimer's disease across different ancestries, including African ancestry, although its effect can vary between groups. The main point is that if a score cannot properly track an important disease-related variant in a population, its performance will be poor. Better genetic markers and ancestry-specific data are needed to improve the score.

5. Limitations

The study used simulated disease data and small group sizes (347–500 people), with no confidence intervals. It tested only one score and one disease. The five-bin calibration results may be noisy, and only age was included as an additional factor. Population structure and relatedness were not adjusted for. Also, the broad 1000 Genomes ancestry groups do not show the genetic differences within each population.

Reference

Zhang, Q., et al. (2020). Risk prediction of late-onset Alzheimer's disease implies an oligogenic architecture. *Nature Communications, 11*, 4799. https://www.nature.com/articles/s41467-020-18534-1 (PGS Catalog PGS000334)